Exponentials And Logarithms

Starter

Solution 2

1. Evaluate when x = 3 and y = −1.

a. 5^x — a 125

b. 3^y — b $\frac{1}{3}$

c. 2^{2x-1} — c 32

d. 7^{1-y} — d 49

e. 11^{x+3y} — e 1

2. Write as a single power.

a. $6^8/6^2$ — a 6^6

b. $y^3(y^9)^2$ — b y^{21}

c. $2^5 \times 2^9/2^8$ — c 2^6

d. $\sqrt{x^8}$ — d x^4

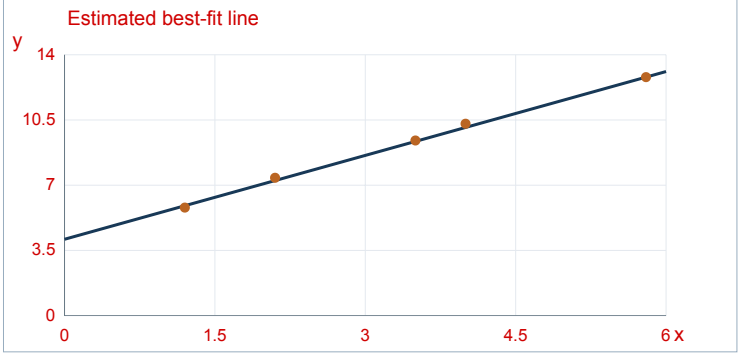
3. Sketch a scatter graph and estimate a straight-line model.

x	1.2	2.1	3.5	4	5.8
y	5.8	7.4	9.4	10.3	12.8

3: model

$y \approx 1.5x + 4.1$

3: graph



Chapter Overview

Solution 3

Exponential functions

Recognise exponential growth and decay, sketch graphs and differentiate exponential functions.

Logarithms

Use inverse functions, logarithm laws and natural logarithms to solve equations.

Modelling

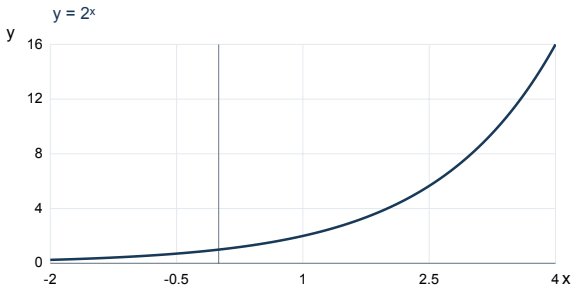
Interpret model parameters and use logarithmic transformations to fit straight lines to nonlinear data.

$y=2^x$

Solution 4

$y = 2^x$

x	-2	-1	0	1	2	3	4
y	$1/4$	$1/2$	1	2	4	8	16



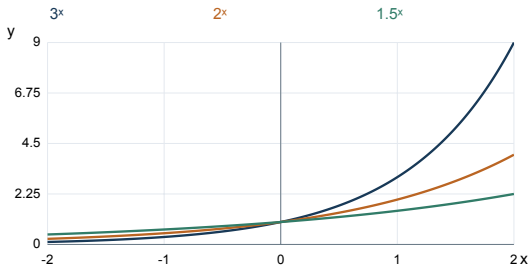
Each increase of 1 in x multiplies y by 2. The graph passes through (0,1), remains positive and approaches y = 0 as x decreases.

Contrasting exponential graphs

Solution 5

Compare $y = 3^x,\ y = 2^x,\ y = 1.5^x$.

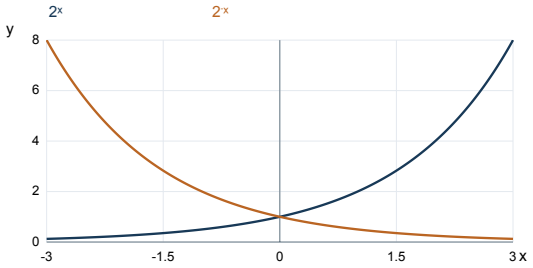
All pass through (0,1). For positive x, the larger base grows faster; for negative x, the ordering reverses.



Compare $y = 2^x$ and $y = 2^{-x}$.

Relationship

Replacing x by −x reflects the graph in the y-axis. The second function decreases, but stays positive.



Graph Transformations

Solution 6

Sketch $y = 2^{x+3}$. Describe its transformation from $y = 2^x$, and state its y-intercept.

Transformation

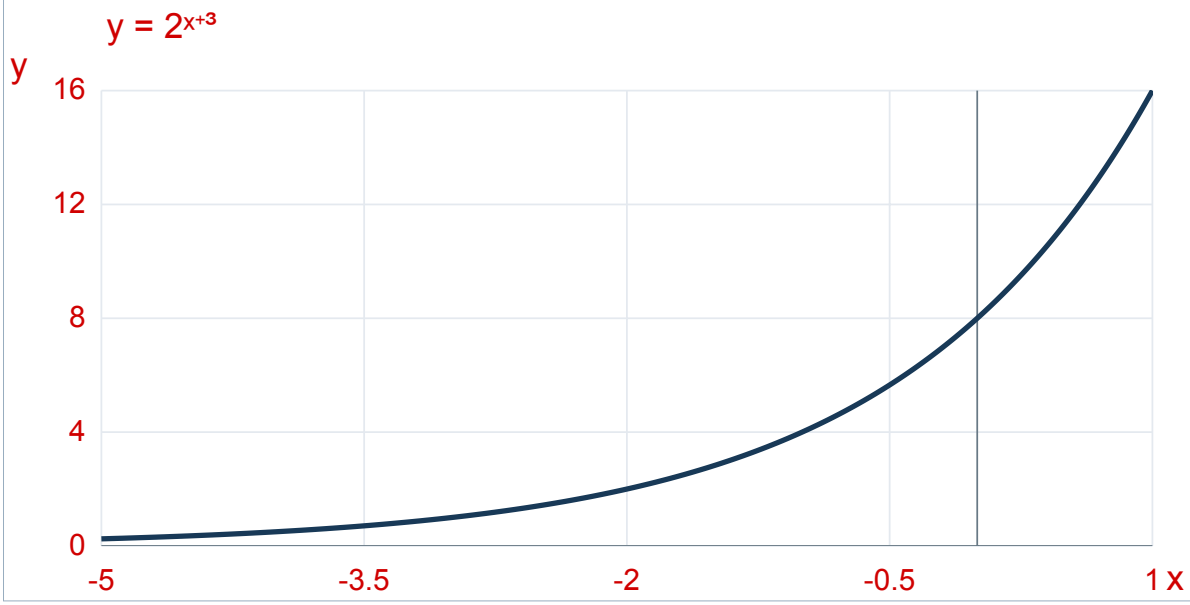
Translate 3 units to the left.

Intercept

$y(0) = 2^3 = 8$

The y-intercept is (0,8); the horizontal asymptote is y = 0.

Sketch



Exercise 14A (Pages 313-314) — Solution 7

6. The curve $y = ka^x$ passes through (1,6) and (4,48). Find k and a.

6

$ka = 6, \quad ka^4 = 48 \Rightarrow a^3 = 8$

$a = 2, \quad k = 3, \quad y = 3(2^x)$

7. The curve $y = pq^x$ passes through $(-3,150)$ and (2,0.048). Find p and q and sketch the curve.

7

$pq^{-3} = 150, \quad pq^2 = 0.048$

$q^5 = 0.00032 \Rightarrow q = 0.2, \quad p = 1.2$

The curve decreases through (0,1.2), approaching $y = 0$.

$y=e^x$ — Solution 8

The gradient of an exponential is proportional to its value.

y	dy/dx (approximately)
1^x	0
1.5^x	$0.41(1.5^x)$
2^x	$0.69(2^x)$
2.5^x	$0.92(2.5^x)$
3^x	$1.10(3^x)$
3.5^x	$1.25(3.5^x)$

Is there a base for which the multiplier is exactly 1?

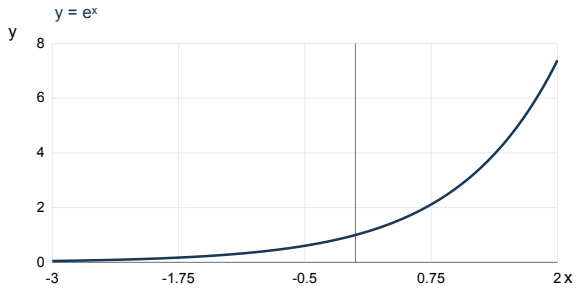
Special base

Yes. It lies between 2.5 and 3 and is called e.

$y=e^x$ — Solution 9

$e = 2.718281828\ldots$

The exponential function with base e has gradient equal to its own value.



$$y = e^x \quad \Longrightarrow \quad \frac{dy}{dx} = e^x$$

It passes through (0,1), where its gradient is 1, and has horizontal asymptote $y = 0$.

Differentiating $y=ae^{kx}$ — Solution 10

Differentiating an exponential

$$\frac{d}{dx}(ae^{kx}) = ake^{kx}$$

$y = e^{5x}$

$y\prime = 5e^{5x}$

$y = e^{-x}$

$y\prime = -e^{-x}$

$y = 4e^{3x}$

$y\prime = 12e^{3x}$

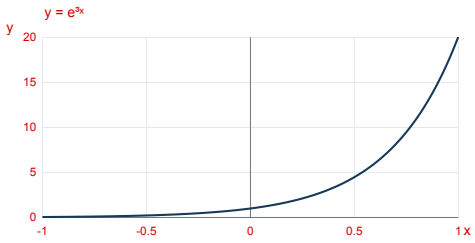
More Graph Transformations

Solution 11

Sketch $y = e^{3x}$.

Increasing; y-intercept (0,1); horizontal asymptote y = 0.

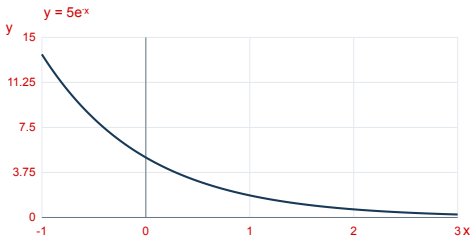
1: sketch



Sketch $y = 5e^{-x}$.

Decreasing; y-intercept (0,5); horizontal asymptote y = 0.

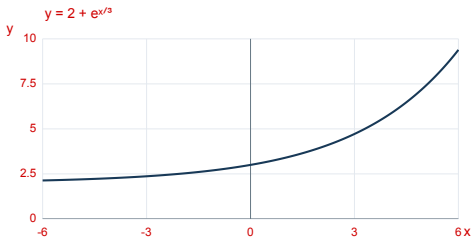
2: sketch



Sketch $y = 2 + e^{x/3}$.

Increasing; y-intercept (0,3); horizontal asymptote y = 2.

3: sketch



Test Your Understanding

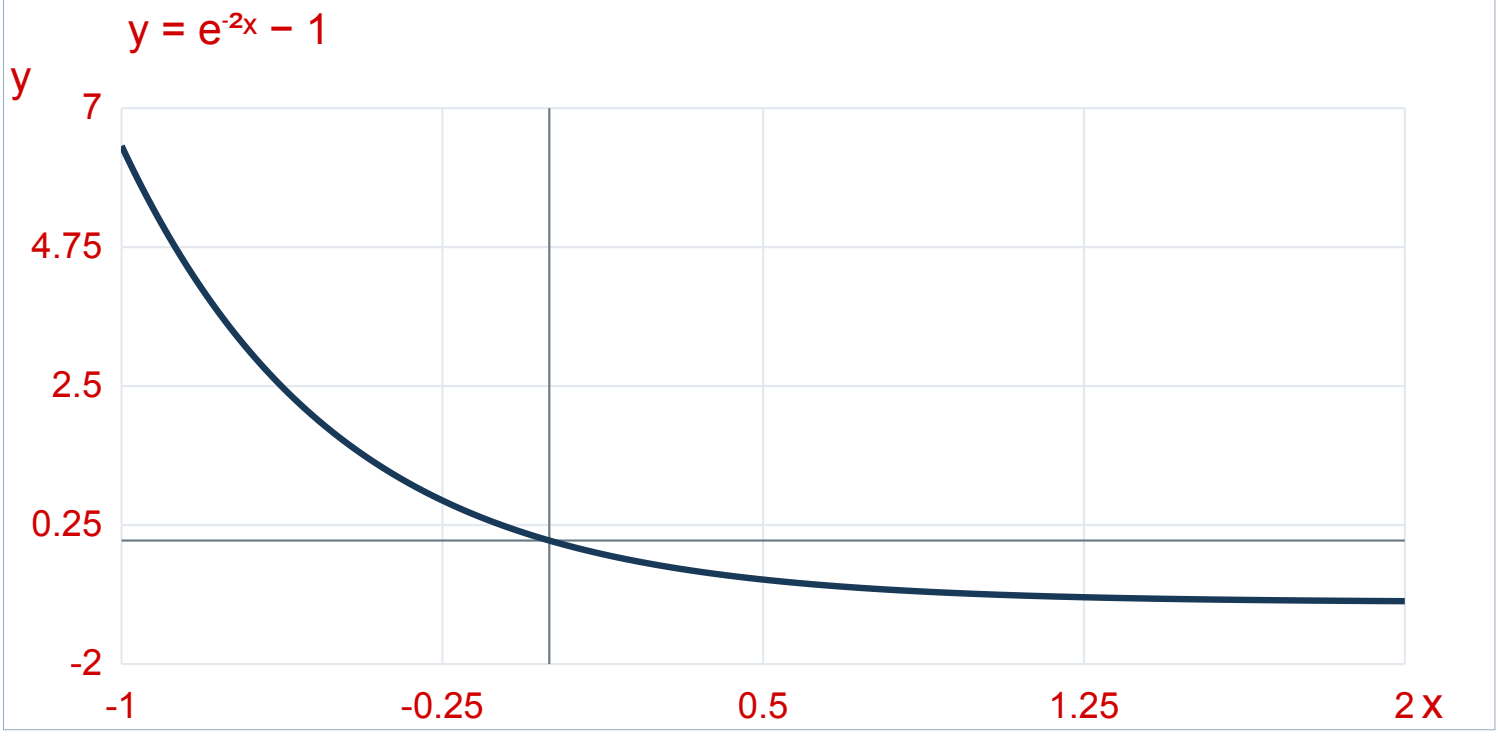
Solution 12

Sketch $y = e^{-2x} - 1$, stating its intercept and asymptote.

Sketch features

The curve decreases through (0,0). It approaches y = −1 as x increases.

Sketch



Exercise 14B (Pages 316-317) Solution 13

Each graph has the form $y = Ae^{bx} + C$. Find A and C, and state the sign of b.

(a) Increasing; horizontal asymptote y = 5; y-intercept (0,6).

(b) Decreasing; horizontal asymptote y = 0; y-intercept (0,4).

(c) Increasing; horizontal asymptote y = 2; y-intercept (0,8).

a

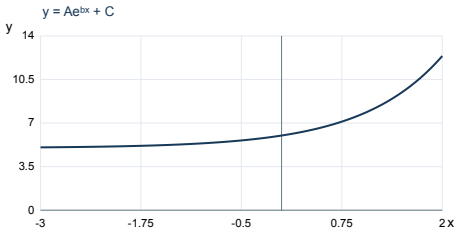
$A = 1,\ C = 5,\ b > 0$

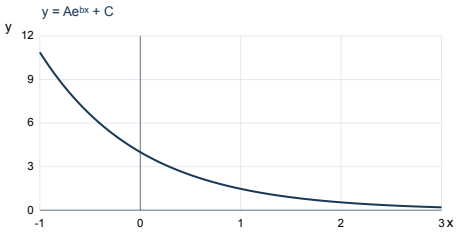
b

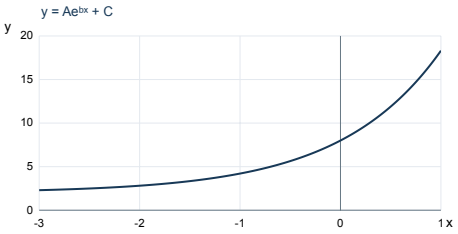
$A = 4,\ C = 0,\ b < 0$

c

$A = 6,\ C = 2,\ b > 0$







Exercise 14B (Pages 316-317) Solution 14

Differentiate each function.

1. e^{6x}

1

$6e^{6x}$

2. $e^{-x/3}$

2

$-\frac{1}{3}e^{-x/3}$

3. $7e^{2x}$

3

$14e^{2x}$

4. $5e^{0.4x}$

4

$2e^{0.4x}$

5. $e^{3x} + 2e^{x}$

5

$3e^{3x} + 2e^{x}$

6. $e^{x}(e^{x} + 1)$

6

$2e^{2x} + e^{x}$

Exercise 14B (Pages 316-317) Solution 15

Find the gradient of $y = e^{3x}$ at x = 2, 0 and −0.5.

Gradients

$y\prime = 3e^{3x}$

$3e^{6}, \quad 3, \quad 3e^{-1.5}$

Find the tangent to $y = e^{0.2x}$ at (5,e).

Tangent

$m = 0.2e, \quad y - e = \frac{e}{5}(x - 5)$

$y = \frac{e}{5}x$

Just for your interest... Solution 16

Where does e arise?

Invest £1 at a nominal annual interest rate of 100%, compounded in n equal instalments.

Balance after one year $= \left(1 + \frac{1}{n}\right)^{n}$

n	1	2	12	365	10000
Balance	2	2.25	2.6130	2.7146	2.7181

Limit

$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^{n} = e$

Exponential Modelling

Solution 17

A population is modelled by $P = 100e^{3t}$. Find its initial size and initial rate of growth.

Initial size

$P(0) = 100$

Initial rate

$\frac{dP}{dt} = 300e^{3t} \Rightarrow P\prime(0) = 300$

An exponential model multiplies by the same factor over equal time intervals. Its instantaneous rate of change is proportional to its current value.

Another Example

Solution 18

Pesticide density is $P = 160e^{-0.006t}$ mg/m², t days after application.

(a) Estimate P after 15 days.

(b) Interpret 160.

a $P(15) = 160e^{-0.09} \approx 146.2 \text{ mg/m}^2$

b The initial density is 160 mg/m².

(c) Show that $dP/dt = kP$.

(d) Interpret the sign of k. (e) Sketch P against t.

c

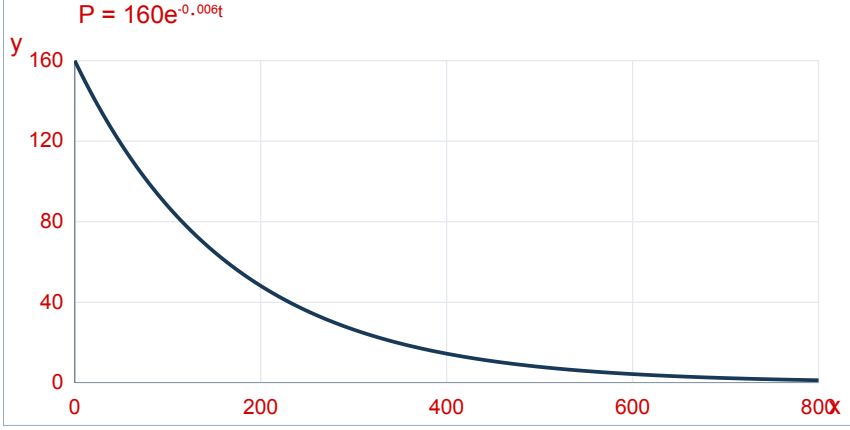
$\frac{dP}{dt} = -0.006(160e^{-0.006t}) = -0.006P$

$k = -0.006$

d–e

The negative sign represents decay. The curve starts at (0,160), decreases and approaches P = 0.

Graph



Exercise 14C (Pages 318-319) Solution 19

Air pressure at height h km is modelled by $p = e^{-0.13h}$ atmospheres.

(a) Estimate pressure at 4.394 km.

a

$p = e^{-0.13(4.394)} \approx 0.565$

(b) Express dp/dh in terms of p.

b

$\frac{dp}{dh} = -0.13p$

(c) Explain how pressure changes with height.

c

It decreases at a rate proportional to the current pressure.

(d) Find the percentage decrease for each additional kilometre.

d

$\frac{p(h+1)}{p(h)} = e^{-0.13}$

$100(1 - e^{-0.13}) \approx 12.2\%$

Exercise 14C (Pages 318-319) Solution 20

A £20,000 tractor depreciates according to two proposed models, t years after purchase.

$T_1 = 20000e^{-0.24t}, \qquad T_2 = 19000e^{-0.255t} + 1000$

(a) Compare their values after one year.

a

$T_1(1) \approx £15733, \quad T_2(1) \approx £15723$

The predictions differ by about £10.

(b) Compare their values after ten years.

b

$T_1(10) \approx £1814, \quad T_2(10) \approx £2484$

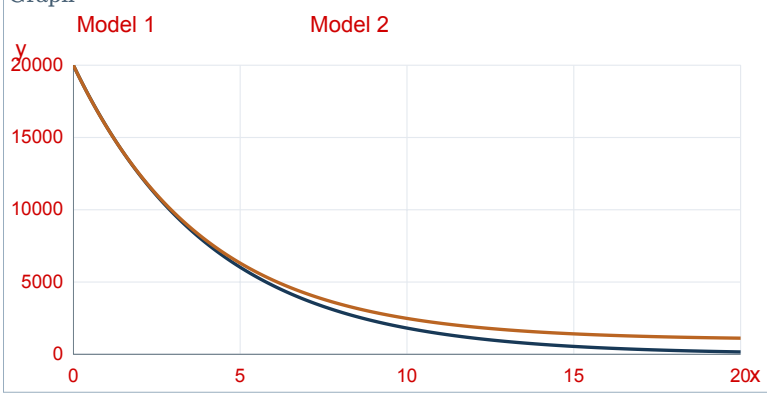
Model 2 predicts the larger value.

(c) Sketch both models. (d) Interpret the 1000 in model 2.

c–d

Both decrease from £20,000. Model 1 approaches zero; model 2 approaches £1,000, which could represent scrap value.

Graph



Logarithms

Solution 21

Logarithms undo exponentiation.

$a^x = n \iff \log_a n = x$

For real logarithms: $a > 0,\ a \neq 1,\ n > 0$.

Operation	Inverse operation
$3x, \quad x+3, \quad x^2\ (x \geq 0), \quad x^5, \quad 3^x$	Inverses $x/3, \quad x-3, \quad \sqrt{x}, \quad \sqrt[5]{x}, \quad \log_3 x$

Interchanging between exponential and log form

Solution 22

$\log_a n = x$ (said "log base a of n") is equivalent to $a^x = n$.
The logarithm outputs the missing power. Here $a > 0$, $a \neq 1$ and $n > 0$.

$$3^2 = 9 \iff \log_3 9 = 2$$

Here are two methods of interchanging between these forms. Pick your favourite!

Method 1: Missing power

The base of the logarithm must match the base of the exponential.
$\log_2 8$ asks: "2 to what power gives 8?"

Imagine inserting the output of the logarithm just after its base. Click the button to move the missing power.

$$\log_2 8 = 3$$

$2^3 = 8$

Method 2: Do the same operation to each side

Since KS3, you have used the same operation on each side of an equation to undo whatever you want to remove.
For example, subtract 2 from each side:
$3x + 2 = 11 \quad \Rightarrow \quad 3x = 9$.

"Log base a" and "a to the power of" undo each other: they are inverse functions.

$\log_2 8 = 3$
Apply $2^{\square}$ to each side:
$2^{\log_2 8} = 2^3 \quad \Rightarrow \quad 8 = 2^3$.

Examples

Solution 23

Evaluate each logarithm by finding the missing power.

1. $\log_5 25$

1
2

2. $\log_3 81$

2
4

3. $\log_4 1$

3
0

4. $\log_2 32$

4
5

5. $\log_4 4$

5
1

6. $\log_2(1/2)$

6
-1

7. $\log_3(1/27)$

7
-3

8. $\log_2(1/16)$

8
-4

9. $\log_4(-1)$

9
undefined over the reals

10. $\log_{10} 1000$

10
3

11. $\log_a(a^3)$

11
3

With Your Calculator

Solution 24

Use a calculator. Give numerical answers to 4 significant figures.

1. $\log_3 7$

1
1.771

2. $\log_5 0.3$

2
-0.7481

3. $\ln 10$

3
2.303

4. $\ln e$

4
1

5. $\log_{10} 100$

5
2

Exercise 14D – Y1 Page 321 — Solution 25

6(a). Without a calculator, justify why $\log_2 50$ lies between 5 and 6. (b) Evaluate it to 4 significant figures.

6

$2^5 = 32 < 50 < 64 = 2^6$

$5 < \log_2 50 < 6, \quad \log_2 50 \approx 5.644$

7(a). Evaluate $\log_2 2,\ \log_3 3,\ \log_{17} 17$. (b) Explain the general rule.

7

$1,\ 1,\ 1$

Since $a^1 = a$, $\log_a a = 1$ for a > 0, a ≠ 1.

8(a). Evaluate $\log_2 1,\ \log_3 1,\ \log_{17} 1$. (b) Explain the general rule.

8

$0,\ 0,\ 0$

Since $a^0 = 1$, $\log_a 1 = 0$ for a > 0, a ≠ 1.

Extension Exercise — Solution 26

1. [MAT 2015] Without a calculator, identify the largest number.

$$A = \frac{\sqrt{7}}{2},\ B = \frac{5}{4},\ C = \frac{\sqrt{10!}}{3(6!)}$$

$$D = \frac{\log_2 30}{\log_3 85},\ E = \frac{1+\sqrt{6}}{3}$$

1

$A^2 = \frac{7}{4},\ B^2 = \frac{25}{16},\ C^2 = \frac{7}{9}$

Since $\log_2 30 < 5$ and $\log_3 85 > 4$, D < 5/4 < A.

Also

$E^2 = \frac{7+2\sqrt{6}}{9} < \frac{13}{9} < \frac{7}{4}$

Therefore A is largest.

2. [MAT 2013] Positive a, b, c satisfy $\log_b a = 2$, $\log_b(c-3) = 3$, $\log_a(c+5) = 2$. Is a uniquely determined?

2

$a = b^2,\ c-3 = b^3,\ c+5 = a^2$

$b^3(b-1) = 8$

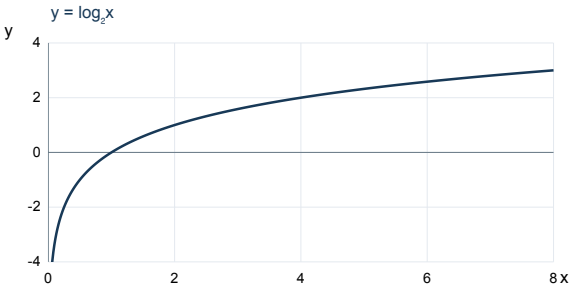
For 0 < b < 1 the left side is negative. For b > 1 it is strictly increasing, so there is exactly one solution.

$b = 2, \quad a = 4, \quad c = 11$

$y = \log_2 x$ — Solution 27

$y = \log_2 x$

x	$1/4$	$1/2$	1	2	4	8
y	-2	-1	0	1	2	3



The domain is x > 0; the range is all real numbers. The graph increases through (1,0) and has vertical asymptote x = 0. It is the reflection of y = 2^x in y = x.

Laws of Logs — Solution 28

Logarithm laws

For a valid base a and positive u, v:

$\log_a(uv) = \log_a u + \log_a v$

$\log_a(u/v) = \log_a u - \log_a v$

$\log_a(u^k) = k \log_a u$

Change of base

$$\log_a u = \frac{\log_b u}{\log_b a}$$

In particular, $\log_a 1 = 0$ and $\log_a a = 1$.

Examples

Solution 29

Combine into a single logarithm (1–4); expand using log laws (5–8). Assume positive variables and valid bases.

1. $\log_3 6 + \log_3 7$

1
$\log_3 42$

2. $\log_2 15 - \log_2 3$

2
$\log_2 5$

3. $2\log_5 3 + 3\log_5 2$

3
$\log_5 72$

4. $\log_{10} 3 - 4\log_{10} 0.5$

4
$\log_{10} 48$

5. $\log_a(x^2yz^3)$

5
$2\log_a x + \log_a y + 3\log_a z$

6. $\log_a(x/y^3)$

6
$\log_a x - 3\log_a y$

7. $\log_a(x\sqrt{y}/z)$

7
$\log_a x + \frac{1}{2}\log_a y - \log_a z$

8. $\log_a(x/a^4)$

8
$\log_a x - 4$

Anti Laws

Solution 30

These are NOT logarithm laws.

$\log_a(u+v) \neq \log_a u + \log_a v$ in general

Counterexample

$\log_{10}(10+10) = \log_{10} 20 \neq 2$

$\log_{10} 10 + \log_{10} 10 = 2$

$(\log_2 x)^3 \neq 3\log_2 x$ in general

Distinction

At x = 2, the left side is 1 and the right side is 3. The power rule applies to a power INSIDE the logarithm:

$\log_2(x^3) = 3\log_2 x$

Solving Equations with Logs

Solution 31

Solve $\log_{10} 4 + 2\log_{10} x = 2$.

$\log_{10}(4x^2) = 2 \Rightarrow 4x^2 = 100$

$x^2 = 25 \Rightarrow x = \pm 5$

The original logarithm requires x > 0. Therefore x = 5.

Test Your Understanding

Solution 32

Solve $2\log_2(x+15) - \log_2 x = 6$.

$\log_2 \dfrac{(x+15)^2}{x} = 6$

$(x+15)^2 = 64x$

2

$x^2 - 34x + 225 = (x-9)(x-25) = 0$

$x = 9 \quad \text{or} \quad x = 25$

Both satisfy the domain x > 0.

Exercise 14E (323-324) Solution 33

Solve $\log_3(x+1) = 1 + 2\log_3(x-1)$.

$\log_3(x+1) = \log_3[3(x-1)^2]$

$x+1 = 3(x-1)^2$

$3x^2 - 7x + 2 = (3x-1)(x-2) = 0$

$x = \frac{1}{3}$ or $x = 2$

The domain is x > 1, so only x = 2 is valid.

Exercise 14E (323-324) Solution 34

Given $a + b = 13$, $\log_6 a + \log_6 b = 2$ and a > b, find a and b.

Answer

$ab = 6^2 = 36, \quad a(13-a) = 36$

$(a-9)(a-4) = 0 \Rightarrow a = 9,\ b = 4$

Challenge: prove the quotient law for logarithms.

Proof

Let $m = \log_a u$ and $n = \log_a v$. Then

$u = a^m,\ v = a^n,\ \frac{u}{v} = a^{m-n}$

$\log_a(u/v) = m - n = \log_a u - \log_a v$

Extension Exercises

Solution 35

1. Solve $\log_3(x-7) - \frac{1}{2}\log_3 x = 1 - \log_3 2$.

1

$$\frac{x-7}{\sqrt{x}} = \frac{3}{2} \Rightarrow 2x - 14 = 3\sqrt{x}$$

Put u = $\sqrt{x} > \sqrt{7}$:

$$2u^2 - 3u - 14 = (2u-7)(u+2) = 0$$

$$u = \frac{7}{2} \Rightarrow x = \frac{49}{4}$$

2. Given $(\log_3 p)^2 = \log_3(p^2)$ and $\log_3(p+q) = \log_3 p + \log_3 q$, find p and q.

2

Put u = $\log_3$p. Then $u^2 = 2u$, so p = 1 or 9. The second equation gives p + q = pq. The value p = 1 is impossible.

$$p = 9, \quad q = \frac{9}{8}$$

3. If $4(\log_{10} a)^2 + (\log_{10} b)^2 = 1$, find the greatest a.

3

$$|\log_{10} a| \le \tfrac{1}{2} \Rightarrow a \le \sqrt{10}$$

Equality occurs at b = 1.

4. Prove $3/2 < \log_2 3 < 5/3$.

4

$$3^2 > 2^3 \Rightarrow 2\log_2 3 > 3$$

$$3^3 < 2^5 \Rightarrow 3\log_2 3 < 5$$

Solving equations with exponential terms

Solution 36

Solve using logarithms; give answers to 3 decimal places.

1. $3^x = 20$

1

$$x = \frac{\ln 20}{\ln 3} \approx 2.727$$

2. $5^{4x-1} = 61$

2

$$x = \frac{1 + \ln 61/\ln 5}{4} \approx 0.889$$

Solving equations with exponential terms

Solution 37

Solve $3^x = 2^{x+1}$.

Take logarithms to any one consistent base.

$x \ln 3 = (x+1) \ln 2$

$x(\ln 3 - \ln 2) = \ln 2$

$$x = \frac{\ln 2}{\ln 3 - \ln 2} \approx 1.710$$

Test Your Understanding

Solution 38

Solve each exponential equation.

1. $3^{2x-1} = 5$

2. $2^x 3^{x+1} = 5$

1

$$x = \frac{1 + \ln 5 / \ln 3}{2} \approx 1.232$$

2

$$x = \frac{\ln(5/3)}{\ln 6}$$

3. $3^{x+1} = 4^{x-1}$

3

$$x = \frac{\ln 12}{\ln(4/3)} \approx 8.638$$

Exercise 14F – Y1 Page 325 — Solution 39

3(a). Solve $3^{x+1} = 2000$.

3a

$$x = \frac{\ln 2000}{\ln 3} - 1 \approx 5.92$$

3(b). Solve $\log_5(x-3) = -1$.

3b $x - 3 = 5^{-1} \Rightarrow x = 3.2$

4(a). Sketch $y = 4^x$. 4(b). Solve $4^{2x} - 10(4^x) + 16 = 0$.

4a

The graph increases through (0,1), with asymptote $y = 0$.

4b

$$u = 4^x > 0, \quad (u-2)(u-8) = 0$$

$$x = \frac{1}{2} \quad \text{or} \quad x = \frac{3}{2}$$

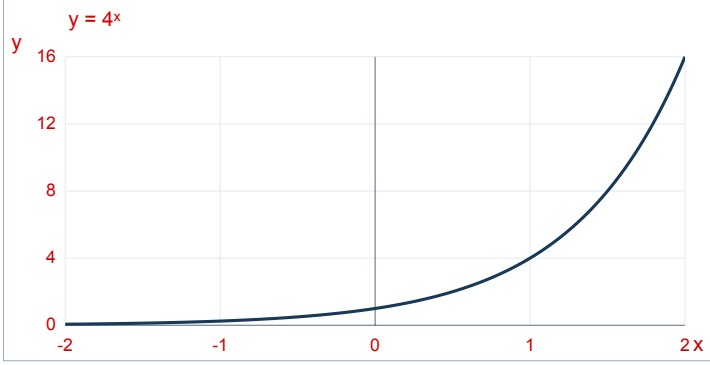
Extension [MAT 2011]: how many positive solutions satisfy $x = 8^{\log_2 x} - 9^{\log_3 x} - 4^{\log_2 x} + 2$?

Extension

$$x = x^3 - 2x^2 + 2 \Rightarrow (x+1)(x-1)(x-2) = 0$$

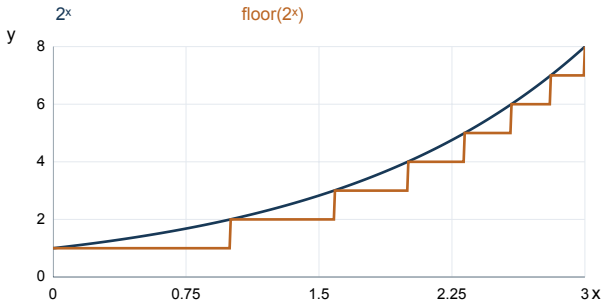
There are two positive solutions: x = 1 and x = 2.

Graph



Very Challenging Extension! — Solution 40

[MAT 2013] Write $\lfloor x \rfloor$ for the greatest integer $\leq$ x. For a natural number n, evaluate $\int_0^n \lfloor 2^x \rfloor \, dx$.



On $\log_2 k \leq x < \log_2(k+1)$, the height is k. Put $N = 2^n$.

Rectangles

$$I = \sum_{k=1}^{N-1} k[\log_2(k+1) - \log_2 k]$$

Simplify

$$I = (N-1)\log_2 N - \sum_{k=1}^{N-1} \log_2 k$$

$$= n(N-1) - \log_2((N-1)!)$$

$$= n2^n - \log_2((2^n)!)$$

Natural Logarithms

Solution 41

The natural logarithm is $\ln x = \log_e x$. It is the inverse of e^x.

$\ln(e^x) = x, \qquad e^{\ln x} = x \ (x > 0)$

Solve $e^x = 5$.

$x = \ln 5$

Solve $2 \ln x + 1 = 5$.

2

$\ln x = 2 \Rightarrow x = e^2$

Quadratics in e^x

Solution 42

Solve $e^{2x} + 2e^x - 15 = 0$.

$(e^x + 5)(e^x - 3) = 0$

Since $e^x > 0$, reject $e^x = -5$.

$e^x = 3 \Rightarrow x = \ln 3$

Solve $e^x - 2e^{-x} = 1$.

$e^{2x} - e^x - 2 = 0$

$(e^x + 1)(e^x - 2) = 0$

Reject $e^x = -1$.

$x = \ln 2$

A substitution $u = e^x$ turns each equation into a quadratic. Remember that u must be positive.

Test Your Understanding

Solution 43

Solve exactly.

1. $\ln(3x+1) = 2$

1

$$x = \frac{e^2 - 1}{3}$$

2. $e^{2x} + 5e^x = 6$

2

$$(e^x + 6)(e^x - 1) = 0 \Rightarrow x = 0$$

3. $2^x e^{x+1} = 3$

3

$$x \ln 2 + x + 1 = \ln 3 \Rightarrow x = \frac{\ln 3 - 1}{\ln 2 + 1}$$

Exercise 14G – Y1 Page 327

Solution 44

4. Solve $e^x + 12e^{-x} = 7$ exactly.

4

$$e^{2x} - 7e^x + 12 = (e^x - 3)(e^x - 4) = 0$$

$$x = \ln 3 \quad \text{or} \quad x = \ln 4$$

6. Solve $3^x e^{4x-1} = 5$ exactly.

6

$$x \ln 3 + 4x - 1 = \ln 5$$

$$x = \frac{1 + \ln 5}{4 + \ln 3}$$

Exercise 14G – Y1 Page 328

Solution 45

8. The graph is $y = 3 + \ln(4 - x)$. Find its exact intercepts A on the y-axis and B on the x-axis.

Challenge: $g(x) = Ae^{Bx} + C$ passes through (0,5) and (6,10), with asymptote y = 2. Find B.

a $A = (0, 3 + \ln 4)$

b

$\ln(4 - x) = -3 \Rightarrow x = 4 - e^{-3}$

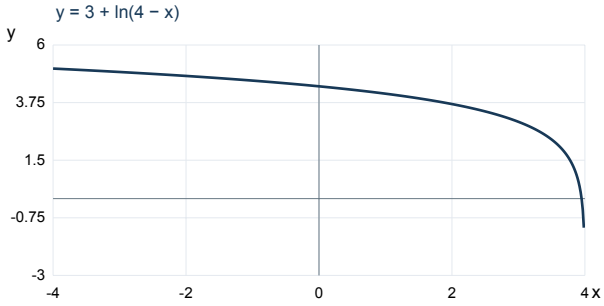
$B = (4 - e^{-3}, 0)$

Challenge

$C = 2, \quad A = 3$

$3e^{6B} + 2 = 10 \Rightarrow e^{6B} = \frac{8}{3}$

$B = \frac{1}{6} \ln \frac{8}{3}$

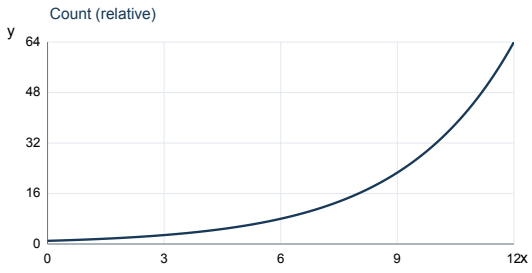


Graphs for Exponential Data

Solution 46

Exponential data can span a very wide range.

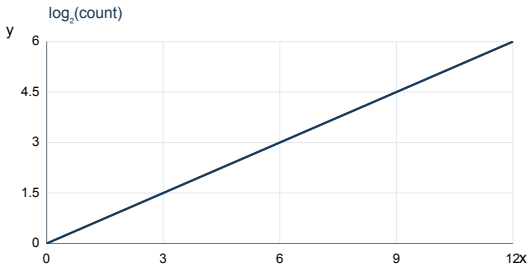
A historical computing model doubles transistor counts every two years. On an ordinary vertical scale, the smaller values cluster near the axis.



Taking logarithms turns multiplication into addition.

$\log_2(2y) = 1 + \log_2 y$

Every doubling adds 1 to the logged value. Plotting $\log_2$(count) against time gives a straight line under this model.



Graphs for Exponential Data

Solution 47

Logarithmic scales

A base-10 logarithmic scale compresses large ratios into manageable differences.

Earthquake amplitude

If the measured amplitude increases by a factor of 10, its base-10 logarithm increases by 1.

$\log_{10}(10A) = 1 + \log_{10} A$

Do not confuse amplitude with energy.

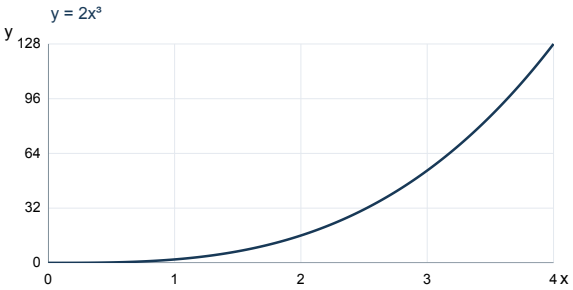
The logarithmic amplitude comparison is not an energy comparison: these are different physical quantities.

Other Non-Linear Growth

Solution 48

Power-law growth: $y = 2x^3$, for x $>$ 0.

A plot on ordinary axes curves upwards. Taking logarithms of BOTH variables produces a straight line.



$\log y = \log 2 + 3 \log x$

Plot log y vertically against log x horizontally: gradient 3 and intercept log 2.

Turning non-linear graphs into linear ones

Solution 49

Power-law model

$y = ax^n \Rightarrow \log y = n \log x + \log a$

Plot log y against log x. Gradient = n; intercept = log a. Require a > 0 and x > 0.

Exponential model

$y = ab^x \Rightarrow \log y = x \log b + \log a$

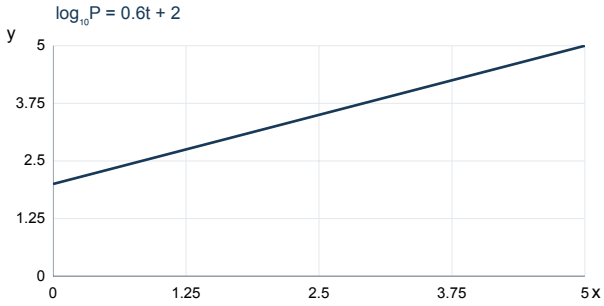
Plot log y against x. Gradient = log b; intercept = log a. Require a > 0 and b > 0.

Use one consistent logarithm base. Compare your transformed equation with Y = mX + c rather than memorising separate rules.

Example

Solution 50

A bacteria population has model $P = ab^t$. Its graph of $\log_{10}$P against t has gradient 0.6 and vertical intercept 2.



(a) Write the line equation.

(b) Find a and b.

(c) Interpret a.

a

$\log_{10} P = 0.6t + 2$

b

$\log_{10} a = 2, \quad \log_{10} b = 0.6$

$a = 100, \quad b = 10^{0.6} \approx 3.98$

c

The initial population is 100.

Example

Solution 51

Historical city data are modelled by $P = aR^n$, where R is rank and P is population.

R	2	3	4	5	6
P	1000000	730000	620000	530000	480000

(a) Calculate $\log_{10}$R and $\log_{10}$P to 2 decimal places.

(b) Plot log P against log R and draw a best-fit line.

(c) Estimate a and n.

a

$\log R$	0.30	0.48	0.60	0.70	0.78
$\log P$	6.00	5.86	5.79	5.72	5.68

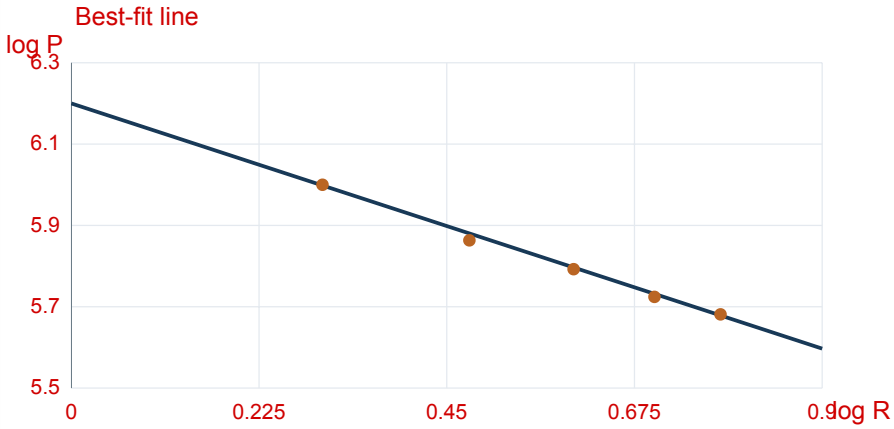
b–c

$\log P \approx -0.67 \log R + 6.2$

$n \approx -0.67, \quad a \approx 10^{6.2} \approx 1.6 \times 10^6$

Graph estimates may vary slightly.

Graph



Test Your Understanding

Solution 52

A website models its followers by $P = ab^t$, t years after the start of 2015.

t	0.7	1.3	2.2
P	2353	3673	7162

(a) Calculate $\log_{10}$P to 3 decimal places.

(b) A best-fit line passes through the first and last transformed points. Its intercept is 3.147. Find its equation.

a

$3.372, \quad 3.565, \quad 3.855$

b

$$m = \frac{3.855 - 3.372}{2.2 - 0.7} = 0.322$$

$\log P = 0.322t + 3.147$

(c) Find the model parameters. (d) Predict P at t = 5.

c–d

$a = 10^{3.147} \approx 1403, \quad b = 10^{0.322} \approx 2.099$

$P(5) = 10^{0.322(5)+3.147} \approx 57148$

Keep unrounded parameters in the prediction.

Fitting model parameters to data is called regression. Logarithmic transformation lets us estimate a nonlinear model using a straight line.

Exercise 14H Page 331-333 — Solution 53

5. Kleiber's law models metabolic rate by $R = am^b$. Mass m is in kg and rate R in kcal/day.

m	0.030	0.408	4.19	34.6	650
R	4.2	32.3	195	760	7637

(a) Calculate $\log_{10}$m and $\log_{10}$R to 2 decimal places.

(b) Plot log R against log m. (c) Estimate a and b from a best-fit line.

a

$\log m$	-1.52	-0.39	0.62	1.54	2.81
$\log R$	0.62	1.51	2.29	2.88	3.88

b–c

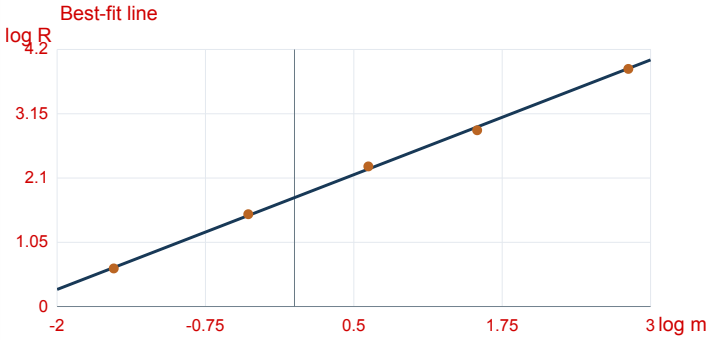
$\log R \approx 0.75 \log m + 1.78$

$a \approx 10^{1.78} \approx 60, \quad b \approx 0.75$

(d) Estimate the metabolic rate for mass 80 kg.

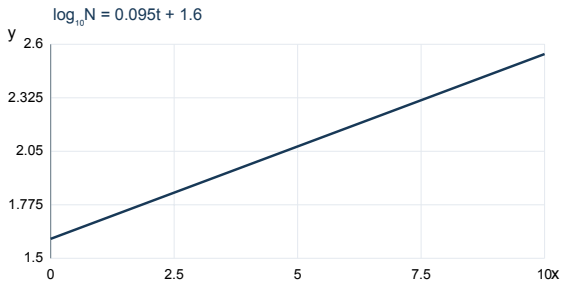
d $R = 60(80)^{0.75} \approx 1600$ kcal/day

Graph



Exercise 14H Page 331-333 — Solution 54

8. A virus model is $N = ab^t$. A graph of $\log_{10}$N against t passes through (0,1.6) and (10,2.55).



(a) Write the line equation.

(b) Find a and b to 2 significant figures.

a $\log_{10} N = 0.095t + 1.6$

b $a = 10^{1.6} \approx 40, \quad b = 10^{0.095} \approx 1.2$

(c) Interpret a.

(d) Predict the number after 30 days using the rounded model in (b).

c Approximately 40 people are sick initially.

d $N = 40(1.2)^{30} \approx 9500$

(e) Why might the model overestimate later numbers?

e People may recover, or transmission may slow. Constant exponential growth will not continue indefinitely.

Rounding matters: retaining the unrounded parameters gives about 28,000 at day 30. Always state which model is used.